

ENVS 193DS Final

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Link to ENV5 193DS Final GitHub Repo: [HERE](#)

Set Up

```
library(janitor)
library(tidyverse)
library(here)
library(readxl)
library(flextable)
library(DHARMA)
library(MuMIn)
library(ggeffects)

# Reading in the data & storing it as an object
nest <- read.csv(here("data", "nest_data_final.csv"))
```

Problem 1: Research Writing

a. Transparents Statistical Methods

Part 1:

In part 1, they more than likely used a logistic regression as the response variable is the probably of American Avocet microhabitat use while the predictor variable is the distance to the water's edge.

Part 2:

In part 2, the they more than likely used a chi-square test of independence as both bird species and habitat type are categorical variables and the analysis looks to examine how the distribution of bird species differs among habitat type.

b. Suggestions for rewriting

Part 1:

American Avocet tuse of micro-habtitats varies wiht distance from the water edge. This suggests that proximity to water influences where and how the American Avocet chooses to use habitats.

Part 2:

American Avocets, Forster's Terns, and Black-necked Stilts used habitat types differently. This suggests that each species has habitat preference differs within the San Francisco Bay estuary.

c. Further Analysis

To further the analysis, examining how the type of vegetation cover within a given area influences American Avocet habitat use. This *continuous variable* could be measured as a percentage (%) of ground cover within the study area using quadrats set at a given dimension (ex: 1m x 1m, 3ft x 3ft) and could showcase how other factors affected by vegetation cover (food, nesting area, protection from predators) influences behavior.

Problem 2: Data Analysis

Goal: Answer the following questions

- 1) How do tree height and acacia ant species influence the probability of bird nest occurrence (of any species of birds)?
- 2) Does acacia ant species identity influence the effect of tree height on probability of nest occurrence differ depending on acacia ant species?

a. Clean & Wrangle

```
# Creating new clean object from nest
nests_clean <- nest |>
  # selecting only the necessary columns
  select("case_control", # selecting case_control
         "height_cm", # selecting bird_species
         "ant_species") |> # selecting ant_species
  # filtering ant_species for the specified species
  filter(ant_species %in% c("RRB", "AB", "BBR")) |>
  # reordering ant species as specified
  mutate(ant_species =
         fct_relevel(ant_species,
                     "AB",
                     "RRB",
                     "BBR")) |>
  # Creating new column with spelled out names
```

```
mutate(ant_species,
      ant_species_name = case_when( # specifying specific names in the new column
        ant_species == "AB" ~ "Crematogaster sjostedti",
        ant_species == "RRB" ~ "Crematogaster mimosae",
        ant_species == "BBR" ~ "Crematogaster nigriceps"
      ))

# Displaying structure of nests_clean
str(nests_clean)

'data.frame': 270 obs. of 4 variables:
 $ case_control : int 1 0 0 0 0 1 0 0 0 1 ...
 $ height_cm : num 392 310 513 154 324 ...
 $ ant_species : Factor w/ 3 levels "AB","RRB","BBR": 2 2 2 2 1 2 3 2 1 3 ...
 $ ant_species_name: chr "Crematogaster mimosae" "Crematogaster mimosae" "Crematogaster mim

# Displaying 10 random rows within the data set
slice_sample(nests_clean, n = 10)
```

	case_control	height_cm	ant_species	ant_species_name
1	0	452.5	RRB	Crematogaster mimosae
2	0	104.0	RRB	Crematogaster mimosae
3	1	181.5	BBR	Crematogaster nigriceps
4	1	254.0	RRB	Crematogaster mimosae
5	0	143.5	RRB	Crematogaster mimosae
6	0	115.5	BBR	Crematogaster nigriceps
7	1	390.0	BBR	Crematogaster nigriceps
8	0	112.0	BBR	Crematogaster nigriceps
9	0	158.0	BBR	Crematogaster nigriceps
10	0	152.0	RRB	Crematogaster mimosae

b. Response Variable

In this data set, “case_control” is made up of a binary system where 1 indicates a bird nest is present in a tree while 0 indicates no bird nests were found within a tree. This is important when assessing how/why birds choose specific trees for nesting.

c. Purpose of the Study

Tree Height:

Tree height is a continuous variable measured in centimeters. As tree height increases, the probability of nest occurrence increases as a taller tree provides greater protection from predators, visibility, and increases structural support to the nest itself (Found in the “Introduction” section).

Ant Species:

Ant species is a discrete categorical variable as it contains three distinct “categories” which in this case are the different species of ants: *Crematogaster sjostedti*, *Crematoogaster mimosae*, and *Crematoogaster nigriceps*. The species of ant can influence nesting probability as different species of ants vary in their overall “aggressiveness” toward other organisms which in turn alters tree structure and can affect the overall suitability of a tree for nesting. (Found in Introduction & Methods).

d. Table of Models

Model number	Tree Height	Ant Species	Interaction	Predictor list
1	X	X	X	All predictors (Tree height and ant species) with an interaction
2	X	X		Tree Height and Ant Species (No Interaction)

e. Run the Models

f. Select the Best Model

```
# Running an AIC
AIC(
  model1,
  model2)
```

```
      df      AIC
model1  6 237.8042
model2  4 258.7430
```

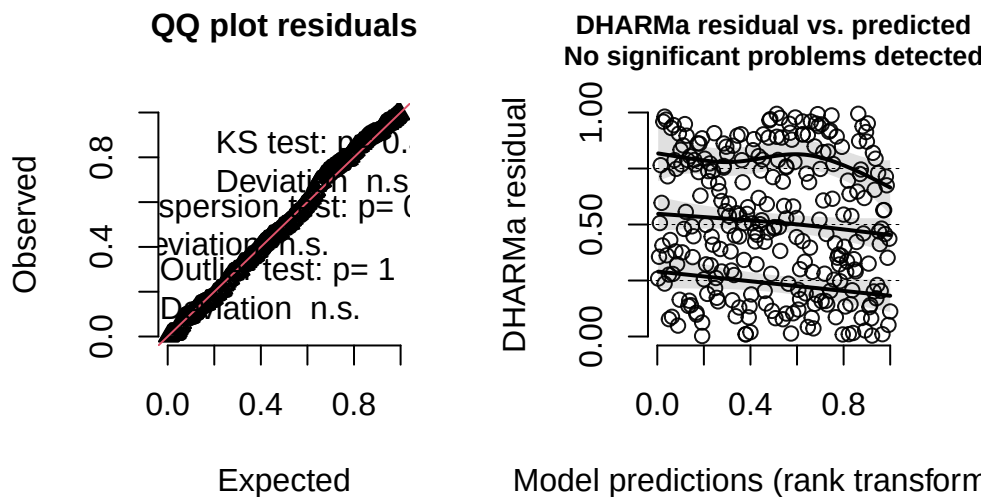
The best model as determined by Akaike's Information Criterion (AIC) included both predictors, tree height (cm) and ant species, and the interaction between both of them (AIC = 237.8). The results suggest that a relationship exists between tree height and the probability of nest occurrence based off the species of acacia ant.

g. Check the Diagnostics

```
# Simulating Residuals
mod1_simres <- simulateResiduals(modell1) # creating new object for simulated residuals

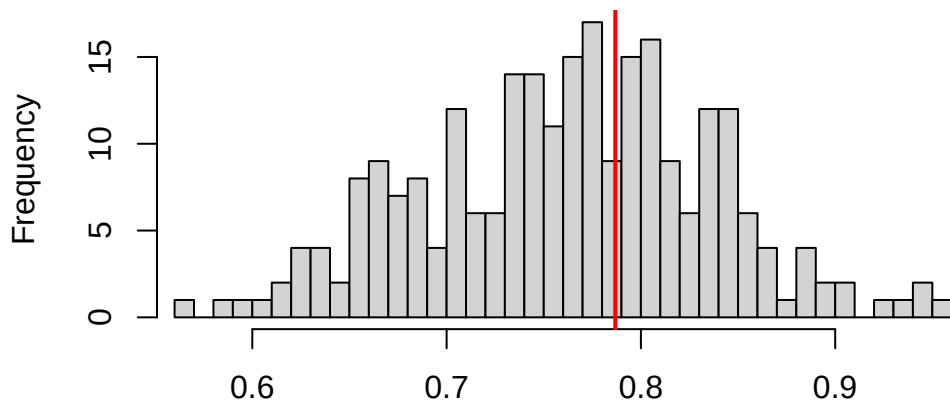
plot(mod1_simres) # plotting the residuals
```

DHARMa residual



```
testDispersion(mod1_simres) # Assessing the dispersion within the model
```

**DHARMA nonparametric dispersion test via sd of
residuals fitted vs. simulated**

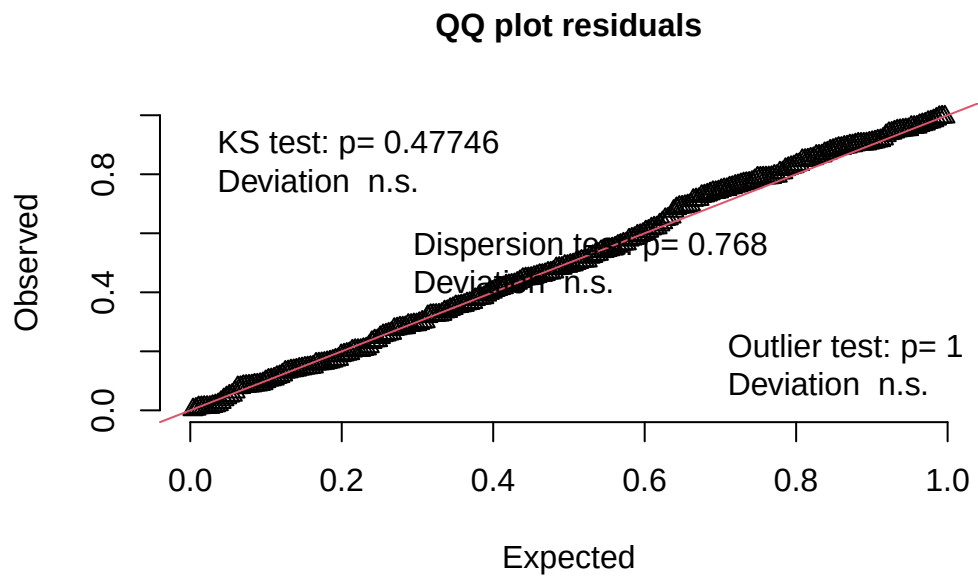


Simulated values, red line = fitted model. p-value (two.sided) = 0.768

DHARMA nonparametric dispersion test via sd of residuals fitted vs.
simulated

```
data: simulationOutput  
dispersion = 1.0325, p-value = 0.768  
alternative hypothesis: two.sided
```

```
testUniformity(mod1_simres) # Assessing the normality of the model
```



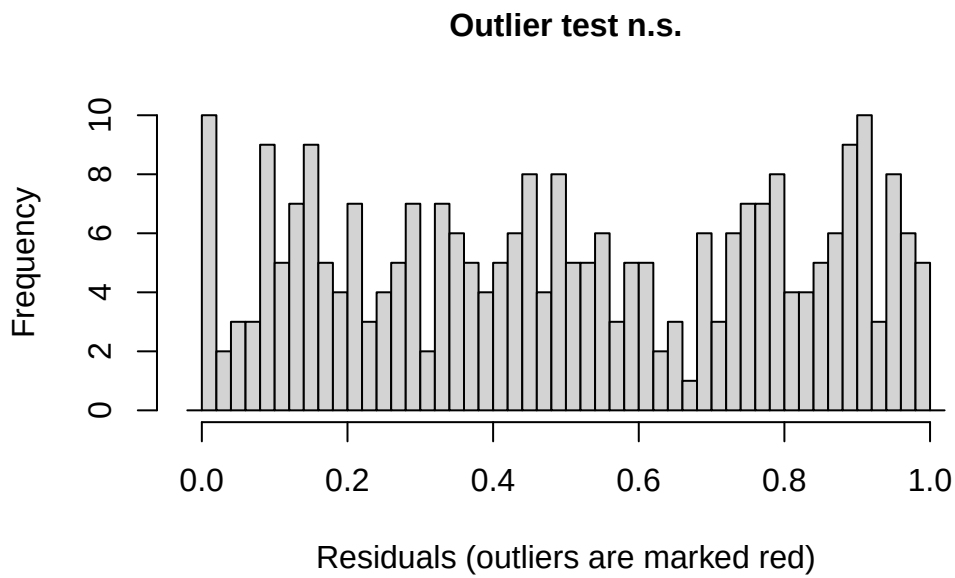
Asymptotic one-sample Kolmogorov-Smirnov test

```
data: simulationOutput$scaledResiduals
```

```
D = 0.051246, p-value = 0.4775
```

```
alternative hypothesis: two-sided
```

```
testOutliers(mod1_simres) # Assessing outliers within the model 1
```

DHARMA bootstrapped outlier test

```
data: mod1_simres
outliers at both margin(s) = 0, observations = 270, p-value = 1
alternative hypothesis: two.sided
percent confidence interval:
 0 0
sample estimates:
outlier frequency (expected: 7.40740740740741e-05 )
0
```

h. Visualize the Model Predictions

i. Write A Figure Caption

j. Calculate Model Predictions

k. Interpret Your Results

Problem 3: Affective and Exploratory Visualizations

a. Comparing Visualizations

b. Designing An Analysis

c. Check Any Assumptions and Run Your Analysis

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Completed during week 10 in workshop!